



University of Wisconsin
SCHOOL OF MEDICINE
AND PUBLIC HEALTH

BMI/STAT-768

Statistical Methods for Medical Image Analysis and High Dimensional Data

Moo K. Chung

Department of Biostatistics and Medical
Informatics

University of Wisconsin-Madison

<https://github.com/lapcebeltrami/BMI768>

Diffusion

Unit Objectives

- 1) Understand diffusion equations
- 2) Understand how to solve them analytically
- 3) Understand how to solve them numerically

Required reading for diffusion on manifold

Chung et al. [arXiv:2203.06665](https://arxiv.org/abs/2203.06665)

Laplace Operator in Euclidean space

The Laplacian measures how a function differs from the average of its neighboring values.

It's the sum of second derivatives — how “curved” the function is at a point!



1 Definition (continuous)

In 2D Euclidean space $(x, y) \in \mathbb{R}^2$,

$$\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

The Laplacian tells us how much f at a point differs from its local average.

2 Finite difference approximation

On a grid with spacing $\delta x = \delta y = 1$, we approximate second derivatives by central differences:

$$\frac{\partial^2 f}{\partial x^2}(x, y) \approx f(x+1, y) - 2f(x, y) + f(x-1, y)$$

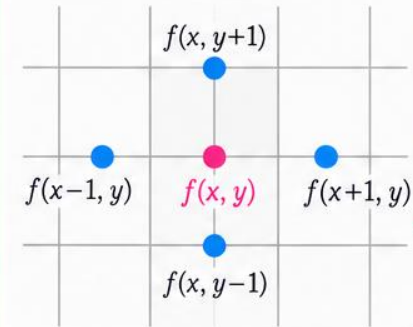
$$\frac{\partial^2 f}{\partial y^2}(x, y) \approx f(x, y+1) - 2f(x, y) + f(x, y-1)$$

Therefore,

$$\Delta f(x, y) \approx f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

3 5-point stencil

The discrete Laplacian uses the center point and its four neighbors (up, down, left, right).



It's like comparing the center value to the average of its neighbors!



4 Matrix form

We can write the discrete Laplacian as

$$\Delta f = \sum_{i,j=-1,0,1} w_{ij} f(x_i, y_j)$$

with weights

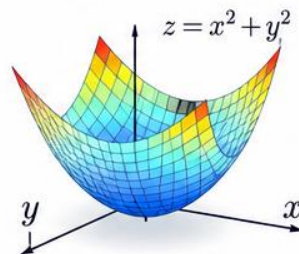
$$(w_{ij}) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

This is called the 5-point Laplacian stencil.

5 Example

If $f(x, y) = x^2 + y^2$, then

$$\Delta f = 2 + 2 = 4.$$



A larger Laplacian means the function is more curved (locally).



6 Higher dimensions

In $\mathbb{R}^n$, the Laplacian is the sum of second partial derivatives:

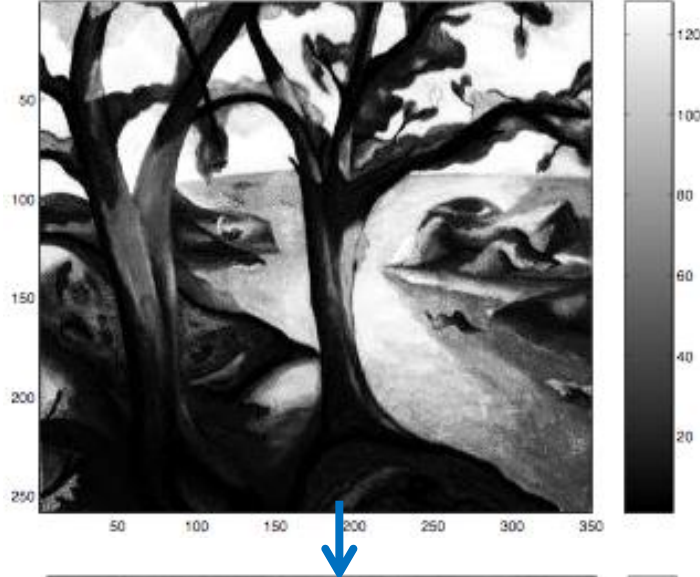
$$\Delta f = \sum_{k=1}^n \frac{\partial^2 f}{\partial x_k^2}$$

On a regular grid with unit spacing, each point has $2n$ neighbors and the center weight is $-2n$.

The Laplacian captures how a function spreads out or deviates from its local average.

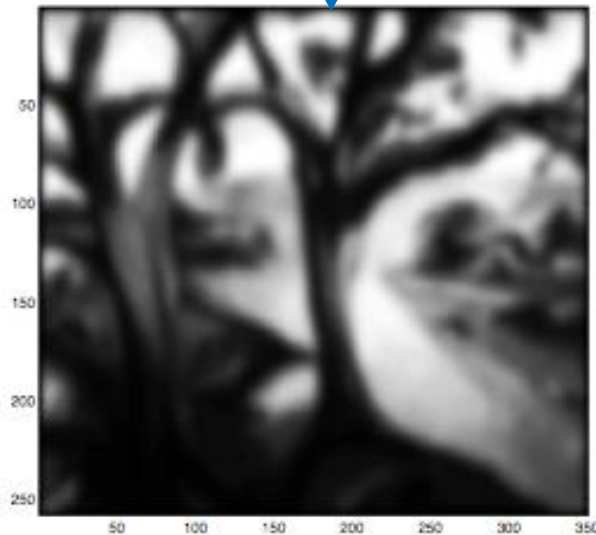
Exercise: Derive graph Laplacian from finite difference

Edge detection using Laplacian

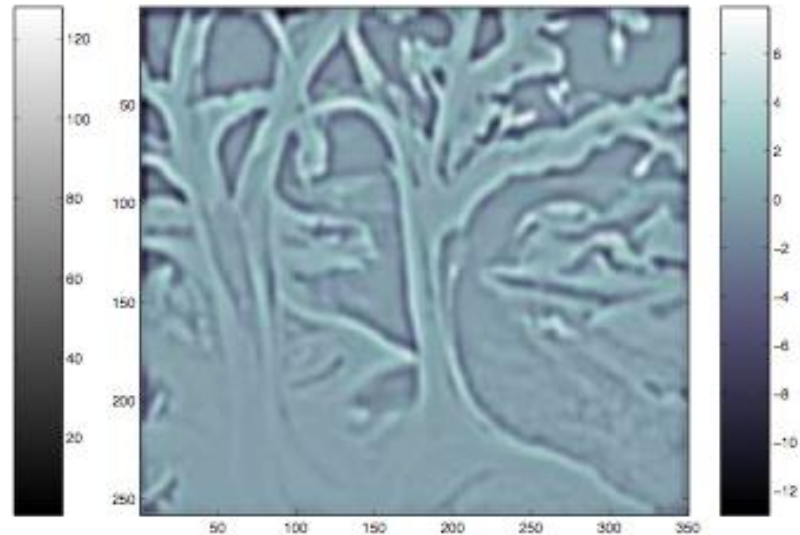


$$(w_{ij}) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\Delta f = \sum_{i,j=-1,0,1} w_{ij} f(x_i, y_i)$$



Gaussian smoothing

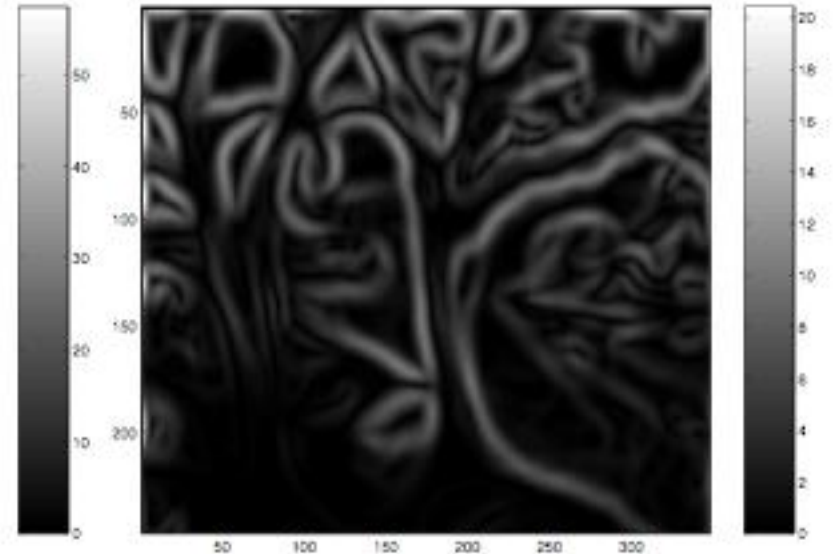
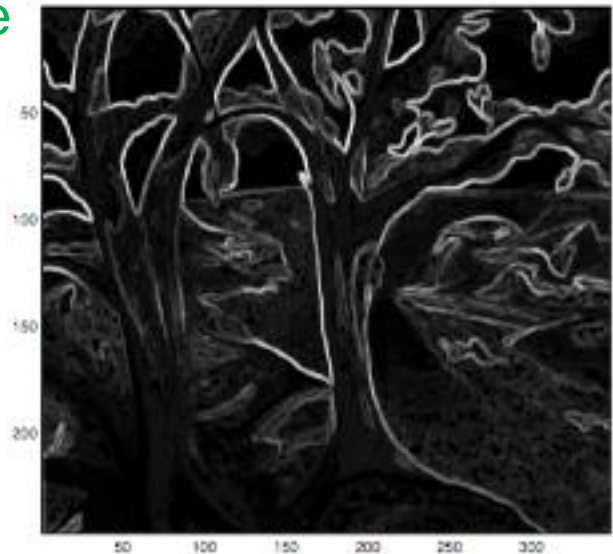
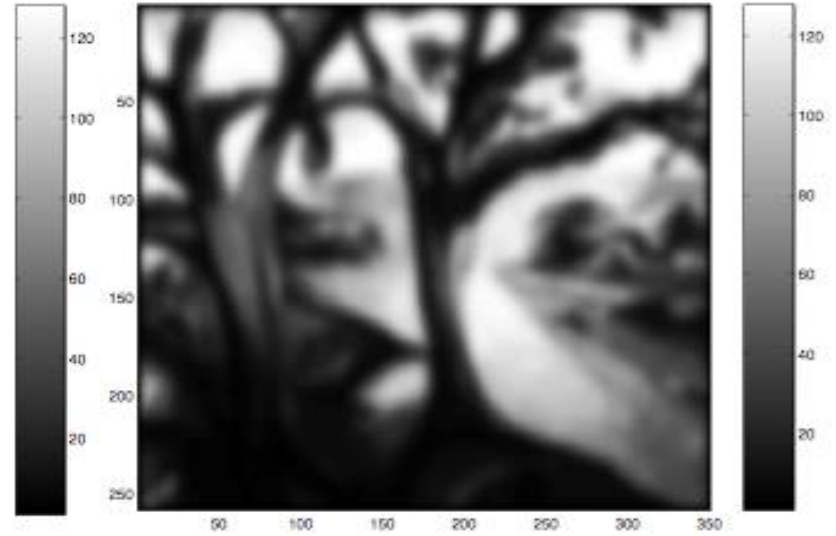
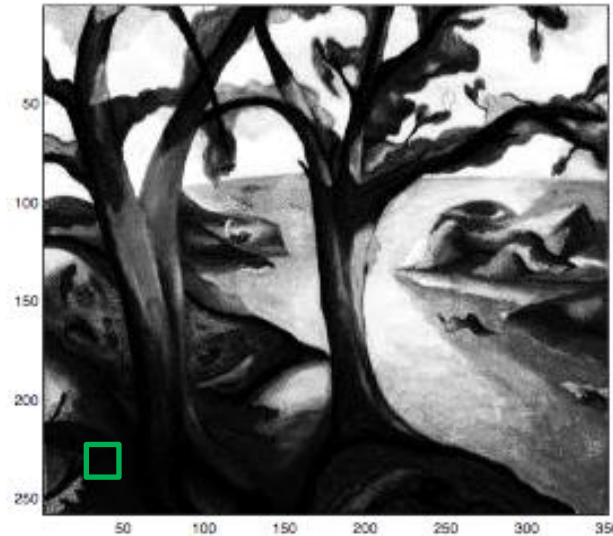


Laplacian

Edge detection using sample variance

Often simple method performs far better than state-of-the-arts

Sample variance within moving window



Without smoothing

With smoothing

1 Diffusion equation in Euclidean space

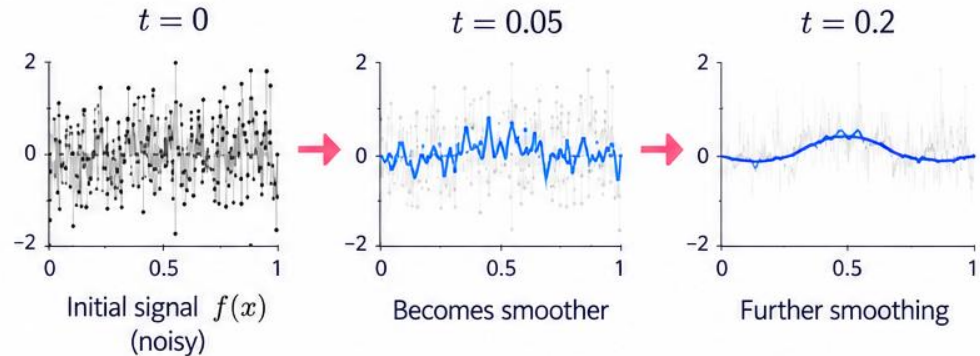
Heat (diffusion) equation:

$$\frac{\partial g(x, t)}{\partial t} = \Delta g(x, t) \quad x \in \mathbb{R}^n, \quad t > 0$$

Initial condition: $g(x, 0) = f(x)$

The function $g(x, t)$ (e.g., temperature or signal) spreads out and becomes smoother over time.

Example: diffusion of a noisy signal (1D)



2 Numerical solution by matrix multiplication

Discretize space on a grid with spacing h (1D):

$$\Delta g(x_i, t) \approx \frac{g_{i+1}(t) - 2g_i(t) + g_{i-1}(t)}{h^2}$$

Write in matrix form:

$$\frac{dg}{dt} = Lg, \quad g(0) = f$$

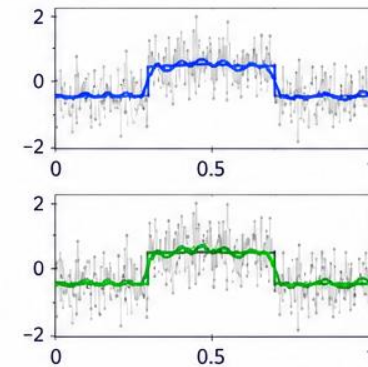
Solve by forward Euler iteration:

$$g^{n+1} = g^n + \Delta t L g^n = (I + \Delta t L) g^n$$

1D Laplacian matrix (N points)

$$L = \frac{1}{h^2} \begin{pmatrix} -2 & 1 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & & 1 & -2 \end{pmatrix}$$

Result



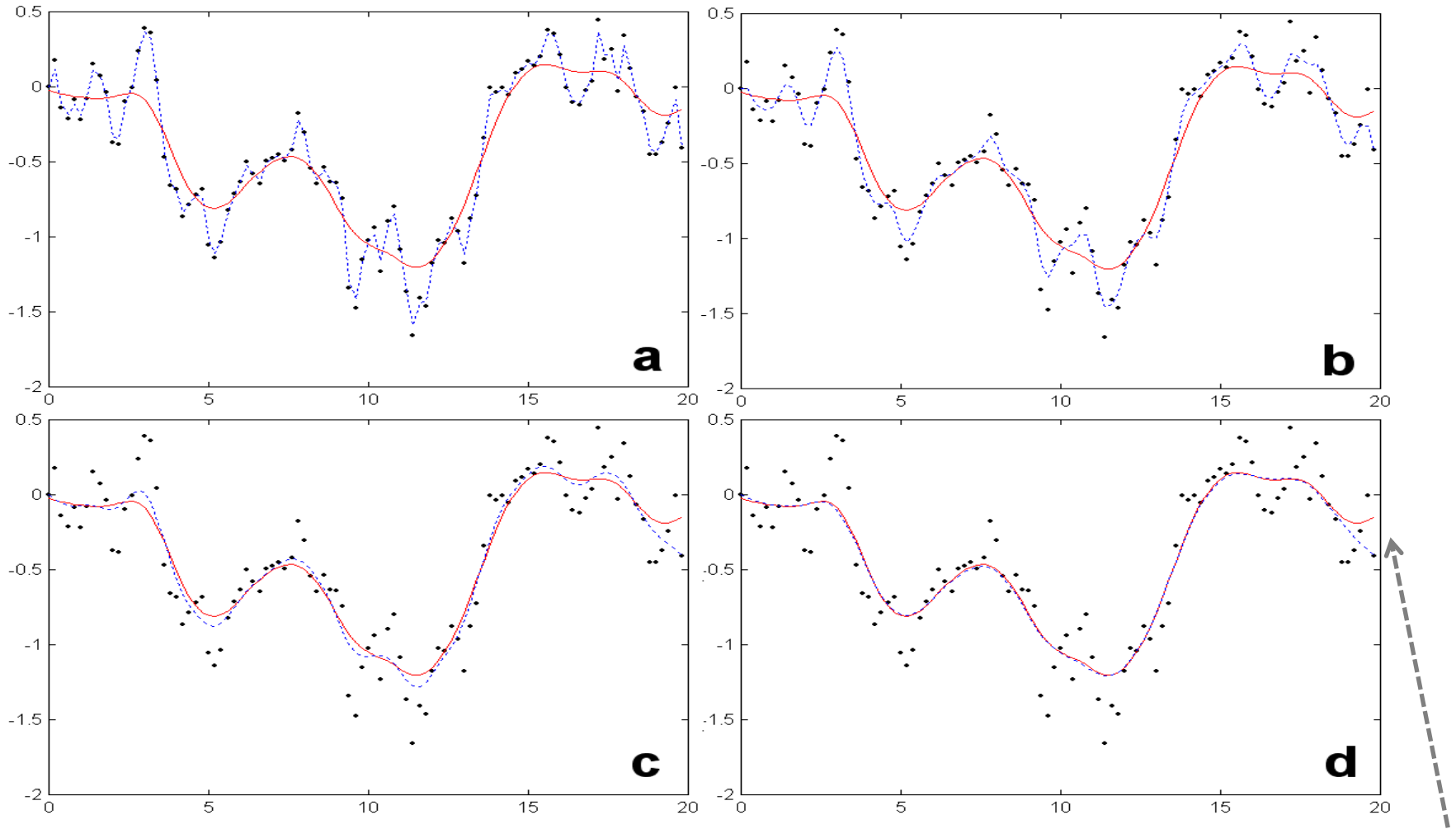
Iterative matrix multiplication result

Toeplitz matrix result

The results exactly match.

Diffusion smooths a function by repeatedly averaging with its neighbors.

Diffusion equation on 1D functional data



Red= Kernel density estimation

Blue = Diffusion equation after 5, 25 and 50 iterations

Shrinkage
at the
boundary

Geometry and Differential Operators on a Surface

All operators are defined using the metric.

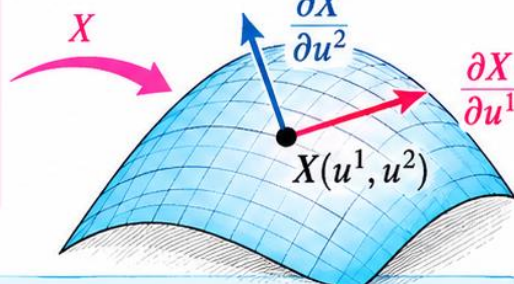
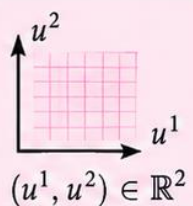
1 Surface and metric tensor

A smooth 2D surface $\mathcal{M}$ is locally parametrized by coordinates (u^1, u^2) in $\mathbb{R}^3$:

$$X(u^1, u^2) = (x(u^1, u^2), y(u^1, u^2), z(u^1, u^2)) \in \mathbb{R}^3.$$

Parameter domain

Surface $\mathcal{M} \subset \mathbb{R}^3$



Metric tensor

The metric tensor measures lengths and angles on the surface. It is the 2×2 matrix

$$g_{ij}(u^1, u^2) = \frac{\partial X}{\partial u^i}(u^1, u^2) \cdot \frac{\partial X}{\partial u^j}(u^1, u^2) \quad (i, j = 1, 2),$$

where $\cdot$ is the usual dot product in $\mathbb{R}^3$.

Explicitly,

$$\begin{aligned} g_{11} &= \frac{\partial X}{\partial u^1} \cdot \frac{\partial X}{\partial u^1}, & g_{12} &= \frac{\partial X}{\partial u^1} \cdot \frac{\partial X}{\partial u^2}, \\ g_{21} &= \frac{\partial X}{\partial u^2} \cdot \frac{\partial X}{\partial u^1}, & g_{22} &= \frac{\partial X}{\partial u^2} \cdot \frac{\partial X}{\partial u^2}. \end{aligned}$$

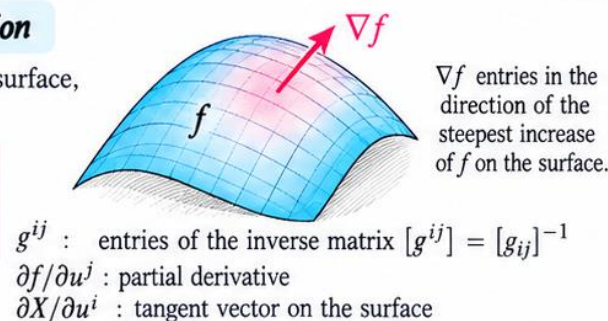
The matrix $[g_{ij}]$ is symmetric and positive definite.

2 Gradient, divergence, and Laplace-Beltrami

2.1 Gradient of a scalar function

For a scalar function $f(u^1, u^2)$ on the surface, the gradient is the tangent vector field

$$\nabla f = \sum_{i=1}^2 \sum_{j=1}^2 g^{ij} \frac{\partial f}{\partial u^j} \frac{\partial X}{\partial u^i},$$

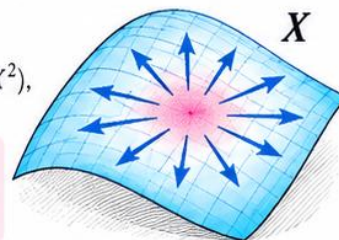


2.2 Divergence of a vector field

For a tangent vector field $X = X^1 \frac{\partial X}{\partial u^1} + X^2 \frac{\partial X}{\partial u^2}$ (with components X^1, X^2), the divergence is the scalar function

$$\nabla \cdot X = \frac{1}{\sqrt{|g|}} \left[\frac{\partial}{\partial u^1} (\sqrt{|g|} X^1) + \frac{\partial}{\partial u^2} (\sqrt{|g|} X^2) \right],$$

where $|g| = \det(g_{ij})$. It measures how much the vector field spreads out (source or sink).

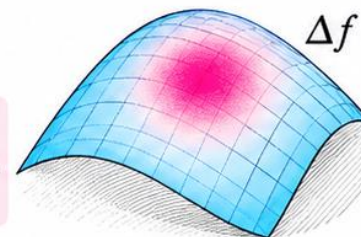


2.3 Laplace-Beltrami operator

For a scalar function f , the Laplace-Beltrami operator is

$$\Delta f = \nabla \cdot (\nabla f) = \frac{1}{\sqrt{|g|}} \sum_{i=1}^2 \frac{\partial}{\partial u^i} \left(\sqrt{|g|} \sum_{j=1}^2 g^{ij} \frac{\partial f}{\partial u^j} \right).$$

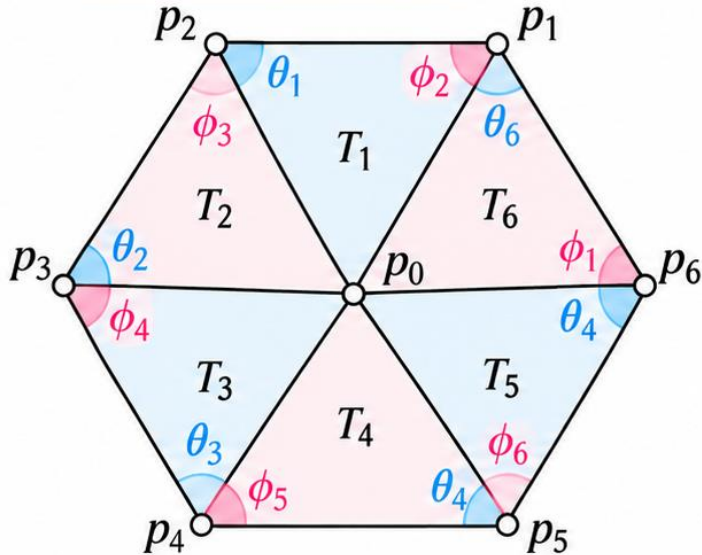
It measures how f differs from the average of its neighbors on the surface. It is the intrinsic generalization of the usual Laplacian.



Δf describes diffusion on the surface.

Cotan discretization of Laplace operator over 2-simplices

1 One-ring around p_0

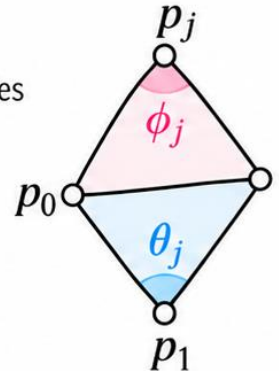


The triangles $T_1, \dots, T_6$ form the one-ring neighborhood of p_0 .

2 Cotan weight

For each edge (p_0, p_j) , let θ_j and ϕ_j be the two angles opposite to the edge (see the adjacent triangles).

$$w_j = \frac{\cot \theta_j + \cot \phi_j}{\sum_{i=1}^m |T_i|}$$



3 Discrete Laplace operator

The cotan discretization of the Laplace operator at vertex p_0 is

$$\Delta f(p_0) = \sum_{i=1}^m w_i [f(p_i) - f(p_0)]$$

4 Key points

- ✓ Uses only local 1-ring neighbors
- ✓ Weights are based on triangle angles (cotangent weights)
- ✓ Widely used in geometry processing and computer graphics

Local geometry $\rightarrow$ Laplace operator

Chung et al. 2003 NeuroImage
Chung et al. 2003 CVPR

Proof: Lopez-Perez et al, 2004, ECCV



diffusion on mesh

Unit Outcomes

- 1) Understand differential operators (DO)
- 2) Learn how to solve DO mathematically
- 3) Learn how to solve DO numerically
- 4) Learn how to set up diffusion equation as matrix multiplication